Evidence for feature-specific error correction in LLMs

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Abstract

Large language models are commonly assumed to use superposition to represent more features than they have dimensions. Theory predicts that computing in superposition requires error correction privileging feature directions over generic ones, but this prediction has only been verified in toy settings. We propose perturbation analysis of residual-stream activations as an empirical test, validated on a toy model with known ground-truth features. We then perturb LLM activations along candidate feature directions from contrastive prompt pairs (pure directions) and along mixtures of two such directions, then compare downstream effects. We find robustness to perturbations in both cases, consistent with error correction, and lower robustness for pure directions, indicating that those candidate directions are privileged by the model; results replicate across Gemma-2-9B, Qwen3-1.7B, Llama-3.1-8B, Mistral-7B-v0.3, Aya-Expanse-8B, and Yi-1.5-9B. We quantify privilegedness by modeling perturbation effect as a function of the L^p-norm of its orthogonal feature components, finding $p > 2$ for contrastive, MELBO, and SAE-decoder directions, and $p \approx 2$ for our random and PCA controls. Since only L^2 is basis-invariant, $p = 2$ means no privileged directions and $p > 2$ is consistent with feature-specific error correction.[1]

1. Introduction

Representations in large language models (LLMs) are poorly understood. It is commonly assumed that LLMs make use of superposition (Elhage et al., 2022) to represent more concepts than they have dimensions available, and potentially to compute in superposition (CiS) (Hänni et al., 2024; Adler & Shavit, 2024; Olah et al., 2025). However, we have only indirect evidence for superposition, chiefly the success of sparse autoencoders (Cunningham et al., 2023; Gao et al., 2024; Templeton, 2024) at extracting interpretable directions, and only theoretical evidence for CiS (Hänni et al., 2024).

An empirical prediction of CiS is that neural networks must correct interference noise while preserving feature signal, which requires sensitivity to perturbations along feature directions over non-feature directions (Hänni et al., 2024). We name this property feature-specific error correction (FSEC). Testing this in LLMs is challenging because we lack ground-truth features. However, FSEC offers a way around this: if the model's error correction privileges certain directions, those directions are candidate features and we can detect them by measuring sensitivity, requiring only a generic input to perturb rather than feature-specific labeled data.

Moreover, finding evidence of FSEC in LLMs would constitute empirical support for CiS.

We provide the first empirical evidence consistent with FSEC, showing that the robustness of LLM activations to perturbations treats feature directions preferentially. Concretely, we perturb residual-stream activations at early layers and measure the downstream response as a function of perturbation direction and magnitude. We construct candidate feature directions via contrastive means for a variety of concepts, including languages, programming languages, gender, sentiment, registers, and verb tenses. Across Gemma-2-9B (Team et al., 2024), Qwen3-1.7B (Yang et al., 2025), Llama-3.1-8B (Grattafiori et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Aya-Expanse-8B (Dang et al., 2024), and Yi-1.5-9B (Young et al., 2024), we find that contrastive feature directions elicit a stronger downstream response than random directions of the same norm, consistent with FSEC that privileges feature directions while suppressing interference along non-feature directions. We formalize this by modeling the LLMs' response to perturbation as an L^p norm of the perturbation's orthogonal feature components. The $p = 2$ case reduces to a basis-invariant quadratic form, meaning no choice of basis is privileged; $p > 2$ breaks this invariance, indicating that the model's response is more sensitive to perturbations along certain directions than others, as predicted by FSEC. We measure $p \approx 2.3$ for con-

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[1]Code: https://anonymous.4open.science/r/fsec-anonymous-5C9D/

trastive directions across models. MELBO (Mack & Turner, 2024b) and SAE directions, which also aim to recover model features, likewise yield $p > 2$, albeit with smaller values. PCA directions and random directions do not, consistent with the interpretation that $p > 2$ reflects alignment with the model's features. We validate this methodology in a toy model with ground-truth features (Section 4.3), confirming that p degrades toward 2 as directions are misaligned with the true features.

Our contributions are:

1. We operationalize an empirical test of FSEC by modeling the LLM's response to perturbation as an L^p norm of orthogonal feature components, where $p > 2$ is a signature of FSEC.
2. We validate our empirical test on a toy model of error correction.
3. We find evidence consistent with FSEC for three types of candidate feature directions (contrastive, SAE-decoder, and MELBO) across six LLMs from different families, with $p > 2$ in each case. PCA and random directions yield $p \approx 2$, consistent with them not being privileged.

2. Related Work

Activation plateaus. Prior work has established that in-distribution activations of LLMs are resistant to perturbations (Heimersheim & Mendel, 2024; Janiak et al., 2024; Shinkle & Heimersheim, 2025), a phenomenon known as *activation plateaus*. We characterize the geometry of the plateau boundary by perturbing along two directions simultaneously, and connect this to error correction.

Error correction for computation in superposition. Hänni et al. (2024) show that error correction is required for computation in superposition in toy models. We provide empirical evidence in favor of superposition and error correction occurring in LLMs.

Direction-dependent sensitivity for feature finding. Prior work has exploited the fact that LLMs have direction-dependent sensitivity to perform unsupervised optimizations for directions maximizing downstream response, resulting in interpretable steering vectors (Mack & Turner, 2024b;a). We make use of the same phenomenon to empirically probe error correction, and apply our analysis to directions obtained in this way (Section 4.2).

3. Methodology

We probe error correction in LLMs by perturbing residual stream activations and measuring the downstream response. This makes use of the fact that language models exhibit direction-dependent sensitivity (Mack & Turner, 2024b;a), which lets us compare the response along candidate feature directions to the response along non-feature baselines.

In all experiments, we perturb the residual stream at an early layer N (default $N = 2$) and measure the downstream response at the second-to-last layer, maximizing the number of intervening layers. This is because the attenuated downstream response phenomenon we observe, *activation plateaus*, is known to be more pronounced the greater the distance between perturbation and measurement (Shinkle & Heimersheim, 2025). We avoid the last residual stream layer, which is known to behave atypically. The downstream response is computed by patching the perturbed activation $\mathbf{a}(\alpha)$ back into the model (Meng et al., 2022; Heimersheim & Nanda, 2024), performing a forward pass, and taking the L^2 distance between the perturbed and unperturbed residual streams at the measurement layer. We show in Section 4.2 that our results are robust to varying both the perturbation and measurement layers, and hold also when measuring cosine distance or KL-divergence in the logits.

To distinguish directional effects from those of magnitude, we perturb by rotating the activation vector $\mathbf{a}$ towards the perturbation direction $\mathbf{d}$ while keeping the activation's norm constant. We refer to this as a *norm-matched perturbation*. Concretely, a perturbation of angle α of $\mathbf{a}$ toward $\mathbf{d}$ is:

$$\mathbf{a}(\alpha) = \cos(\alpha)\mathbf{a} + \sin(\alpha)\,\|\mathbf{a}\|\,\frac{\mathbf{d}_\perp}{\|\mathbf{d}_\perp\|}, \tag{1}$$

where $\mathbf{d}_\perp$ is the component of $\mathbf{d}$ that is orthogonal to $\mathbf{a}$. We typically perturb at the last token position, but show in Section 4.2 that our results are robust to this choice.

We quantify the model's sensitivity along a given direction as the *plateau-breaking angle*, the smallest perturbation angle for which the downstream response exceeds a threshold T. Because absolute response magnitudes differ across models, we set T per-model relative to the population's typical scale. Specifically, for each direction pair k (defined below) we first compute the smaller of the two single-axis maxima, capturing the pair's overall responsiveness without being dominated by one axis. We then take the median of this quantity over all pairs, and scale by a fraction f:

$$T = f \cdot \mathrm{median}_k \min\big(\max_\alpha L_k^2(\alpha, 0),\ \max_\alpha L_k^2(0, \alpha)\big). \tag{2}$$

The median and min ensure that T reflects a typical, attainable response level: without them, a few highly responsive pairs or a single dominant axis within a pair could push T beyond what most pairs ever reach, leaving many plateau-breaking angles undefined. Throughout we use $f = 0.40$, which on Gemma-2-9B yields $T \approx 150$. We show in Section 4.2 that our results are robust to varying T within roughly half-to-double its nominal value.

We perform perturbations along five types of directions: contrastive (Panickssery et al., 2023; Turner et al., 2023), MELBO (Mack & Turner, 2024b), SAE latents (Lieberum et al., 2024), PCA directions, and random. The first three are all candidate feature directions, and PCA and random directions function as non-feature baselines. PCA directions are computed by performing PCA on the residual-stream activations arising from a randomly selected sample of 10000 FineWeb inputs (Penedo et al., 2024). Random directions are sampled isotropically on the unit sphere, $\hat{\mathbf{d}}_{\text{rand}} \sim \text{Unif}(\mathcal{S}^{d-1})$.

Contrastive directions are constructed as the difference of mean activations between two sets of $P \geq 30$ matched prompts $\mathcal{P}^+, \mathcal{P}^-$ differing along a single concept (e.g., for gender, $\mathcal{P}^+ \ni$ "He ran home" is paired with $\mathcal{P}^- \ni$ "She ran home") (Panickssery et al., 2023; Turner et al., 2023). The prompts are LLM generated and human verified. Letting $\mathbf{a}(s) \in \mathbb{R}^d$ denote the activation at the layer of interest for prompt s,

$$\mathbf{d}_{\text{contrast}} = \frac{1}{P}\sum_{i=1}^{P}\big(\mathbf{a}(s_i^+) - \mathbf{a}(s_i^-)\big), \tag{3}$$

$$\hat{\mathbf{d}}_{\text{contrast}} = \mathbf{d}_{\text{contrast}}/\|\mathbf{d}_{\text{contrast}}\|_2. \tag{4}$$

MELBO directions (Mack & Turner, 2024b) are constructed by optimising a unit-norm direction $\hat{\mathbf{d}}$ at the perturbation layer to maximise the L^2 distance between perturbed and unperturbed activations at a downstream layer. Unlike contrastive directions, this requires no labelled prompt pairs as the directions are found unsupervised, and have been shown to produce interpretable steering when applied at sufficient magnitude.

SAE latents are decoder columns of a sparse autoencoder trained on the model's residual stream activations (Bricken et al., 2023; Cunningham et al., 2023). Each column is a candidate feature direction under the SAE's decomposition. We select the top-33 most active SAE latents in the same sample of 10000 FineWeb inputs used for PCA directions. We do this for Gemma-2-9B only and use Gemma Scope (Lieberum et al., 2024).

We also perform perturbations toward combinations of two directions $\mathbf{d}_1, \mathbf{d}_2$. We first restrict them to the tangent space at $\hat{\mathbf{a}} = \mathbf{a}/\|\mathbf{a}\|$ via the projector $I - \hat{\mathbf{a}}\hat{\mathbf{a}}^\top$, and orthonormalize (Gram–Schmidt) to obtain unit vectors $\hat{\mathbf{d}}_1^\perp, \hat{\mathbf{d}}_2^\perp$ satisfying $\hat{\mathbf{a}} \cdot \hat{\mathbf{d}}_i^\perp = 0$ and $\hat{\mathbf{d}}_1^\perp \cdot \hat{\mathbf{d}}_2^\perp = 0$. We then form the unit tangent direction

$$\hat{\mathbf{d}}(\varphi) = \cos\varphi\,\hat{\mathbf{d}}_1^\perp + \sin\varphi\,\hat{\mathbf{d}}_2^\perp, \qquad \varphi \in [0, \pi/2], \tag{5}$$

which interpolates between $\hat{\mathbf{d}}_1^\perp$ at $\varphi = 0$ and $\hat{\mathbf{d}}_2^\perp$ at $\varphi = \pi/2$. Perturbing $\mathbf{a}$ by angle α toward $\hat{\mathbf{d}}(\varphi)$ then proceeds exactly as in the single-direction case,

$$\mathbf{a}(\alpha, \varphi) = \cos\alpha\,\mathbf{a} + \sin\alpha\,\|\mathbf{a}\|\,\hat{\mathbf{d}}(\varphi). \tag{6}$$

We observe behavior (see Section 4) that appears consistent with the downstream response depending on the perturbation's projections onto privileged directions, raised to a common power p. To formalize this, we model the downstream response R to a perturbation vector $\mathbf{v}$ as

$$R(\mathbf{v}) = F\left(\sum_i w_i \left|\hat{\mathbf{f}}_i^\top \mathbf{v}\right|^p\right), \tag{7}$$

where $\{\hat{\mathbf{f}}_i\}$ are unit directions with weights w_i and F is a scalar function.

When we perturb along two directions simultaneously and vary their relative weighting, the resulting plateau boundary traces out a superellipse. Taking $\hat{\mathbf{f}}_1 = \hat{\mathbf{d}}_1^\perp$ and $\hat{\mathbf{f}}_2 = \hat{\mathbf{d}}_2^\perp$, and assuming the remaining $\hat{\mathbf{f}}_j$ contribute negligibly along $\hat{\mathbf{d}}(\varphi)$ over the relevant range of φ, the inner sum collapses to two terms. Under superposition, features are expected to be packed nearly orthogonally (Elhage et al., 2022), so projections of $\hat{\mathbf{d}}(\varphi)$ onto other directions $\hat{\mathbf{f}}_j$ should be small; the quality of the superellipse fits in Section 4 supports this approximation. For a norm-matched perturbation by angle α towards $\hat{\mathbf{d}}(\varphi)$, the perturbation vector is $\mathbf{v} = \sin\alpha\,\|\mathbf{a}\|\,\hat{\mathbf{d}}(\varphi)$, so

$$R = F\big(\sin^p\alpha\,\|\mathbf{a}\|^p\big[w_1\cos^p\varphi + w_2\sin^p\varphi\big]\big). \tag{8}$$

The plateau breaks when F's argument crosses a fixed level ξ^*, giving the implicit equation for the plateau-breaking angle $\alpha(\varphi)$:

$$\sin^p\alpha(\varphi)\,\|\mathbf{a}\|^p\big[w_1\cos^p\varphi + w_2\sin^p\varphi\big] = \xi^*. \tag{9}$$

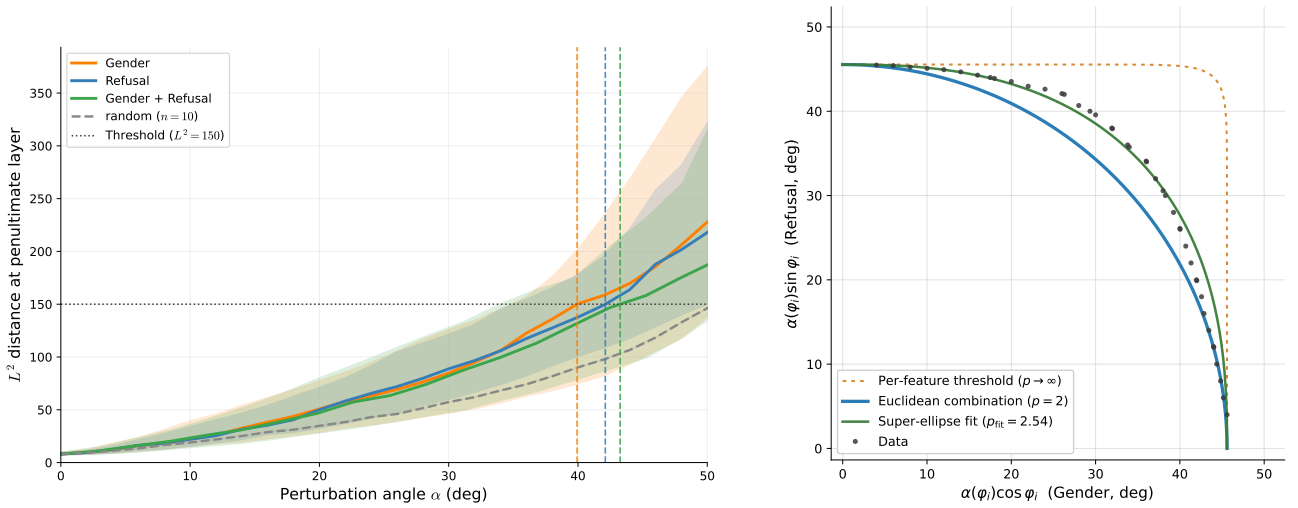
The single-axis sweeps calibrate the weights at this same threshold: $\varphi = 0$ gives $w_1\,\|\mathbf{a}\|^p = \xi^*/\sin^p\alpha_1$, and $\varphi = \pi/2$ gives $w_2\,\|\mathbf{a}\|^p = \xi^*/\sin^p\alpha_2$, where α_1, α_2 are the plateau-breaking angles measured along $\hat{\mathbf{d}}_1^\perp$ and $\hat{\mathbf{d}}_2^\perp$ individually. Substituting and dividing through by ξ^* eliminates the unknown threshold and the activation norm, yielding

$$\left(\frac{\sin\alpha(\varphi)\cos\varphi}{\sin\alpha_1}\right)^p + \left(\frac{\sin\alpha(\varphi)\sin\varphi}{\sin\alpha_2}\right)^p = 1. \tag{10}$$

This means that the plateau-breaking angles define a superellipse of exponent p in the normalised coordinates $(\sin\alpha(\varphi)\cos\varphi/\sin\alpha_1,\ \sin\alpha(\varphi)\sin\varphi/\sin\alpha_2)$. We estimate p by minimizing the squared residual of this equation over the measured $(\sin\alpha(\varphi)\cos\varphi,\ \sin\alpha(\varphi)\sin\varphi)$ points.

4. Results

We begin by illustrating our measurement procedure on a concrete example. Figure 1a shows the downstream L^2



(a) **Measuring plateau-breaking angles.** Perturbation sweeps along the Gender direction, the Refusal direction, and an equal combination of both, at Gemma-2-9B layer 2. The plateau-breaking angle is the angle at which the downstream L^2 distance first exceeds the threshold (here $T = 150$). The grey "Random" curve shows the median (and IQR) response across 10 perturbations along isotropic random unit directions.

(b) **Iso-plateau boundary.** Plateau-breaking angles for the Gender $\times$ Refusal pair at Gemma-2-9B layer 2, threshold $T = 150$. The fitted superellipse exponent is $p_{\text{fit}} = 2.54$, with a fit residual of 0.3%.

Figure 1. **Testing feature-specific error correction.** Left: the downstream response as a function of perturbation angle for two contrastive directions and their combination, illustrating how plateau-breaking angles are extracted. Right: the resulting iso-plateau boundary in normalised coordinates, fit by a superellipse with exponent $p > 2$, indicating that these directions are privileged.

response in Gemma-2-9B when perturbing along the Gender contrastive direction, the Refusal contrastive direction, and an equal combination of the two. The vertical dashed lines mark the plateau-breaking angle for each of the directions. Repeating this measurement across a range of mixing angles φ yields a plateau-breaking angle for each, which we plot in normalised coordinates in Figure 1b. The resulting boundary is well fit by a superellipse with exponent $p_{\text{fit}} = 2.54$.

To verify that combining contrastive directions in this way is meaningful, we show in Figure 2 that steering along these combinations produces interpretable compositional behaviour changes. Concretely, steering with the Gender direction produces feminine-coded completions from a masculine-coded input, steering with the Future Tense direction produces future-tense completions from a past-tense input, and steering along a combination of the two results in feminine-coded, future-tense completions. This supports the reasonableness of the combination-perturbation setup, though we note that interpretable steering alone does not establish that a direction corresponds to a model feature.

4.1. Superellipse exponents across direction types

We now systematically characterise these patterns across directions, direction types, and models. The superellipse exponent p has an intuitive geometric interpretation. If each feature direction has an independent threshold, the plateau boundary traces a square ($p \to \infty$): a downstream response occurs when the projection along either direction exceeds its threshold. If instead the response depends only on the norm of the perturbation, the boundary is an ellipse ($p = 2$), indicating direction-insensitivity. A superellipse of exponent $p > 2$ interpolates between these extremes, with higher p indicating stronger directional privileging.

We repeat the analysis across five direction types. For contrastive directions, we form 318 pairs from 33 directions (14 binary semantic concepts, 10 natural languages, 9 programming languages), filtering for intra-pair cosine similarity below 0.1 (full list and pairwise overlap in Appendix A). For MELBO, we use 528 pairs obtained via the procedure in Section 3. For SAE latents, we use top-activating decoder columns from Gemma Scope's residual stream layer 2 width-16k SAE. PCA and random directions serve as non-feature baselines. We observe that contrastive directions (mean 2.35, 95% CI [2.23, 2.51], median 2.23), MELBO directions (mean 2.23, 95% CI [2.06, 2.40], median 2.19), and top-activating SAE latents (mean 2.18, 95% CI [2.06, 2.33], median 2.10) sit consistently above $p = 2$. This indicates that the model is particularly sensitive to perturbations along these directions, consistent with them being features and with FSEC. The same is not true for either of our baselines: top PCA directions and random directions both have mean and median $p \approx 2$. As discussed in Section 1, $p = 2$ means the plateau boundary is elliptical and the downstream

Prompt:	Once upon a time, the prince
Unsteered:	went out into the world to learn the craft of his father, the king. The prince traveled to the edge of the world, and from there he traveled on to a land where the sun never set. Here he found a wise old man. “What is the highest art of kingship?” asked the prince. “To love and s…
+ Female 20°:	was walking through the woods, and she came across a fairy who asked her what she wanted. And the princess replied, “I want to be loved.” This fairy is very old and has lived in these woods for hundreds of years. She is a very powerful fairy, and when she heard the
+ Future 30°:	was sent to a land where the king is having a problem. The prince will try to solve the problem, but he will fail. So he will go to a tower to get help from a fairy, but he will fail. Then he will go to a tower to get help from a wizard, but he will fail. Finally, he will go…
+ Female 20° + Future 30°:	of an Indian tribe was forced to marry the prince of an African tribe. After they married each other, the queen of the Indian tribe will give birth to two twins. One twin has long hair and one has short hair. The mother of the twin will take her short hair and send her to another

Figure 2. **Compositional steering at Gemma-2-9B layer 2.** Sample completions for the prompt “Once upon a time, the prince” under no steering, +Female alone, +Future alone, and the Female + Future composite. Highlighted spans: blue for Future and orange for Female words.

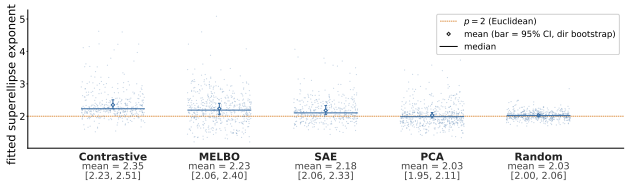


Figure 3. **Superellipse exponents by direction type.** Each dot is one fitted superellipse exponent p for a pair of directions. The black horizontal lines are the per-column medians. The dashed orange line corresponds to the $p = 2$ isotropic reference. The white markers are the per-column means, with error bars corresponding to the 95% confidence interval computed by direction bootstrapping. Candidate feature directions (contrastive, MELBO, top-activating SAE latents) sit consistently above $p = 2$. PCA and random direction baselines cluster at $p \approx 2$. No per-condition median fit residual exceeds 2%.

response breaks through at a perturbation magnitude that depends only on its Euclidean norm, not on its alignment with either direction. This is inconsistent with PCA directions and random directions being aligned with feature directions, under the assumption of FSEC.

4.2. Ablations

We now show that our results hold across a variety of settings. To do so, we vary our setup along the following axes:

- **Perturbation method:** additive rather than norm-matched.
- **Perturbation layer:** $N \in \{5, 10, 20\}$ in addition to the default $N = 2$.
- **Measurement layer:** layers 30 and 35 in addition to the default second-to-last (layer 40 for Gemma-2-9B).
- **Response metric:** cosine distance at second-to-last residual stream layer and KL-divergence in logit space in addition to L^2 distance.
- **Plateau-breaking threshold:** 50% and 200% of the default threshold value.
- **Activation source:** wikipedia-en, wikipedia-zh, and the-stack (default FineWeb).
- **Model:** Qwen3-1.7B, Llama-3.1-8B, Mistral-7B-v0.3, Aya-Expanse-8B, and Yi-1.5-9B.

The results of all these ablations are shown in Figure 4. For contrastive feature directions, the means are all in the [2.15, 2.49] interval, and the medians in [2.10, 2.35], all consistent with contrastive feature directions being privileged.

4.3. Toy model validation

Our candidate feature directions are imperfect approximations of the models’ true features. We expect this to bias the measured p downward, as rotating away from a true feature direction should yield a less privileged direction. We cannot test this in LLMs, where we lack ground-truth features, so we turn to a toy model.

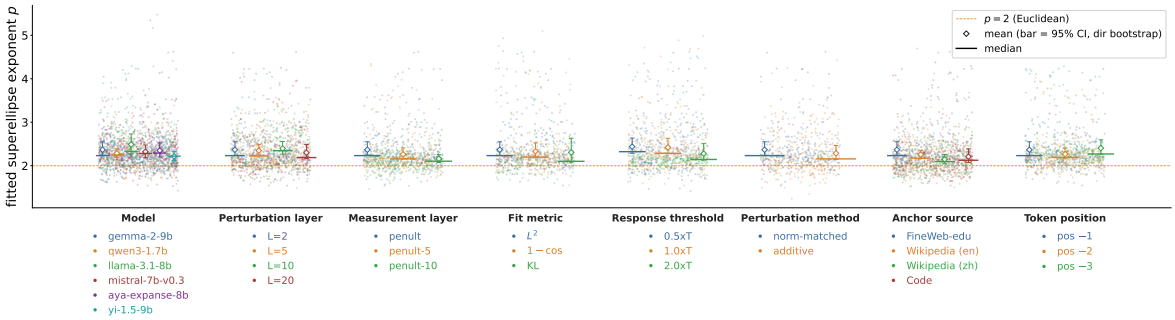


Figure 4. **Robustness of $p > 2$ for contrastive directions across setup choices.** Each dot is one fitted superellipse exponent p for a pair of contrastive directions; the colored marker shows the per-setting mean with 95% CI, and the horizontal dash shows the median. The dashed orange line is the $p = 2$ isotropic reference. Columns vary the model, perturbation layer, response metric, response threshold, perturbation method, activation source, and perturbation token position. Fitted exponents remain above $p = 2$ across all variations. No per-condition median fit residual exceeds 2%.

We use the two-layer denoising network of Hänni et al. (2024); Vaintrob (2026), which denoises two-hot d-dimensional inputs with Gaussian noise using $H < d$ neurons, operating in superposition. Crucially, the input is in the feature basis, giving us access to ground-truth feature directions. We apply our two-direction perturbation analysis as before, perturbing additively along pairs of ground-truth inactive feature directions. To measure the effect of misalignment, we rotate the perturbation directions away from the true feature directions toward random directions by varying amounts. We sweep three feature-to-neuron ratios ($4\times$, $8\times$, $16\times$) with $H = 1024$ neurons. Full details of the toy model setup are in Appendix C.

Figure 5 shows the result. For ground-truth feature directions, we obtain p in the range 3–3.4, well above 2. As directions are rotated away from the true features, p degrades monotonically toward 2, which is recovered for random directions. This is consistent with our prediction that FSEC produces $p > 2$ and that imperfect feature alignment biases p downward. Due to the significant differences between this toy model and LLMs (two layers, binary features, $\tanh^3$ activation function), we do not place much weight on the specific values obtained; we view this primarily as validation of the methodology and the directional privileging prediction.

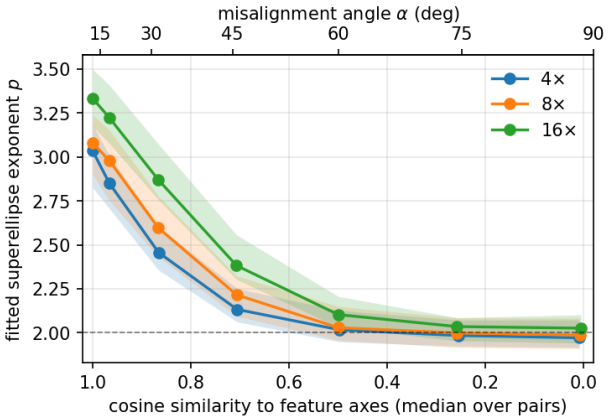


Figure 5. **Superellipse exponents versus alignment with feature directions.** Each dot is the median of 80 fitted superellipse exponents; the shaded region shows the interquartile range. The dashed line marks the $p = 2$ isotropic reference. Perturbation directions take the form $\cos\alpha, \mathbf{e}_j + \sin\alpha, \mathbf{w}$ for $\alpha \in [0, \pi/2]$, where $\mathbf{e}_j$ is a feature axis and $\mathbf{w}$ is an isotropic random unit vector in $\mathbb{R}^d$; the x-axis shows the median realized $|\langle \mathbf{u}, \mathbf{e}_j \rangle|$ over the 80 pairs. The leftmost points correspond to ground-truth feature directions; the rightmost to isotropic random unit vectors, whose realized overlap with the feature axes is of order $1/\sqrt{d}$ per pair. The three lines correspond to feature-to-neuron ratios of $4\times$, $8\times$, and $16\times$, all with $H = 1024$ neurons. Ground-truth feature directions yield p in the range 3–3.4; p degrades monotonically toward 2 as directions are rotated away from the true features.

5. Discussion

5.1. Why $p > 2$ is necessary for error correction

For a neural network to compute in superposition, it must suppress interference noise: when multiple features are active, writing to one feature direction inevitably writes onto others. To distinguish genuine feature activations from interference, the network must treat feature directions differently from non-feature directions. To test whether this is the case, we model the downstream response R to a perturbation vector $\mathbf{v}$ as $R(\mathbf{v}) = F(\sum_i w_i |\hat{\mathbf{f}}_i^\top \mathbf{v}|^p)$ and measure the exponent p.

The exponent p controls how selectively the network can respond to individual feature directions. When $p = 2$, the response reduces to a quadratic form,

$$R(\hat{\mathbf{d}}) = F\left(\hat{\mathbf{d}}^\top \mathbf{A}\, \hat{\mathbf{d}}\right), \tag{11}$$

where $\mathbf{A} = \sum_i w_i \hat{\mathbf{f}}_i \hat{\mathbf{f}}_i^\top$ is a positive semidefinite matrix with at most d_{model} eigenvalues. Under superposition, we

expect $M \gg d_{\text{model}}$ features, yet a quadratic form has only d_{model} degrees of freedom and therefore cannot privilege M directions above all others. Even without superposition, $p = 2$ does not allow all feature directions to be more responsive than all non-feature directions: if the eigenvalues of $\mathbf{A}$ are not all equal, convexity implies that some non-feature directions (linear combinations of high-eigenvalue features) will be more responsive than the lowest-eigenvalue features; if all eigenvalues are equal, the response is isotropic and no direction is privileged at all. In either case, $p = 2$ is insufficient for error correction that privileges feature directions.

At the other extreme, $p \to \infty$ makes the response sensitive only to the single largest feature projection, perfectly suppressing interference from all other directions. It seems unlikely that that is how LLMs function. We therefore expect the true exponent to lie somewhere between these extremes: $2 < p < \infty$.

Empirically, we find $p \approx 2.3$ for contrastive directions and $p \approx 2.2$ for MELBO and SAE directions, consistently above $p = 2$ and consistent with FSEC. Since our candidate feature directions are imperfect approximations of the model's true features, the measured p is likely a lower bound on the exponent that would be obtained with ground-truth feature directions: imperfect alignment would dilute the directional sensitivity, biasing p toward 2. Our toy model analysis (Section 4.3) supports this, showing that p degrades monotonically toward 2 as directions are rotated away from the true features. In fact, moderate misalignment of perturbation directions in the form of ~ 0.8 cosine similarity with the feature directions reduces the fitted superellipse exponent p from 3–3.4 to 2.3–2.6, indicating that our measured $p \approx 2.3$ for contrastive directions in LLMs is consistent with significantly higher p for true feature directions.

5.2. Alternative interpretations

Our finding that $p > 2$ for candidate feature directions is consistent with FSEC. However, there exist alternative interpretations of this result and of activation plateaus more broadly.

Activation plateaus, the near-zero downstream response to small perturbations that underlies our measurement, might not be signatures of error correction at all. They could instead reflect learned robustness, perhaps related to broad basins of attraction in the activation space of LLMs. This could hold whether or not LLMs implement error correction: it could be that error correction is present but plateaus are not the right empirical signature, or that error correction is absent and plateaus reflect a different phenomenon. However, our main result does not depend on interpreting plateaus as error correction per se: even if the mechanism underlying plateaus is not error correction, the finding that $p > 2$ for candidate feature directions but $p \approx 2$ for baselines remains, and still indicates that the network treats these directions preferentially.

It is also possible that LLMs implement error correction in a way that is not directly direction-aware, i.e., without features being linear directions that are consistent across the activation space. For instance, the in-distribution activations of LLMs may lie on a lower-dimensional manifold (Bengio et al., 2017; Cheng et al., 2023), and activation plateaus could reflect a mechanism by which small off-manifold perturbations are mapped back on-manifold. This would be broadly consistent with our observations but would admit a different interpretation than the one we have put forward.

6. Conclusion

We have proposed perturbation analysis of residual-stream activations as an empirical test of whether feature directions are privileged and whether LLMs implement error correction. We find that candidate feature directions in the form of contrastive directions, MELBO directions, and top-activating SAE latents are privileged, consistent with superposition and error correction. Our results hold across six model families and a wide range of experimental settings.

Our results consistently select $p \approx 2.3$ for our best candidates for feature directions. As discussed in Section 5, this is likely a lower bound: imperfect alignment between our candidate feature directions and the model's true features would dilute the directional sensitivity, biasing p toward 2. This provides a quantitative target for toy models of error correction: a good toy model should reproduce $p > 2$ under comparable experimental conditions. Indeed, the toy model of Section 4.3 passes this test, yielding p in the range 3–3.4 for ground-truth features. Extending this analysis, for instance, to standard activation functions or deeper architectures, is a natural direction for future work.

Finally, our method successfully distinguishes candidate feature directions from baselines, which suggests a direction for future work: searching for directions that maximize the superellipse exponent as an unsupervised feature-finding objective.

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A. Contrastive direction inventory

We extract 33 contrastive directions across three families. Each direction is the difference of mean last-token activations at the perturbation layer between a positive prompt set and a matched negative set, normalised to a unit vector.

The 14 **binary semantic** directions are Age, Certainty, Era, Gender, Health, Honesty, Literary, Number, Person, Refusal, Sentiment, Status, Tense, and Wealth. Each uses 30 matched prompt pairs except Literary, which contrasts the slang and literary registers across 50 prompt templates. Prompts are LLM-generated and human-verified.

The 10 **natural-language** directions cover Arabic, Chinese, Dutch, French, German, Italian, Japanese, Portuguese, Russian, and Spanish. Each pair is an English prompt and its translation in the target language; we use 36 matched prompts per direction, drawn from a single multilingual prompt pool so that all 10 directions share the same English source side.

The 9 **programming-language** directions cover C++, Go, Haskell, Java, JavaScript, Lisp, Python, Rust, and TypeScript. Each pair is an English prompt and an equivalent code snippet in the target language; 30 matched prompts per direction.

Forming all $\binom{33}{2} = 528$ unordered pairs, we keep the 318 with raw overlap $|\langle \mathbf{d}_i, \mathbf{d}_j \rangle| < 0.1$ for our two-direction analysis (Figure 6). The filter is needed because for a pair of nearly-parallel directions the orthogonal residual produced by the Gram–Schmidt step (Section 3) is a small fraction of $\mathbf{d}_2$ blown up to unit norm by the renormalisation, and its identity becomes dominated by orthogonalisation noise rather than the labelled feature.

B. Distribution of superellipse fit residuals

For every fitted pair we record the mean radial fraction $\rho = \frac{1}{N}\sum_{k=1}^{N} \left| (|x_k|^p + |y_k|^p)^{1/p} - 1 \right|$, i.e. the average fractional distance of the N contour points from the fitted superellipse, with $\rho = 0$ corresponding to a perfect fit. The number of contour points N varies per fit (it is the cardinality of the iso-threshold level set on the 2D L^2 grid); the per-fit median is 26 (IQR 16–40).

Each pair contributes one dot in Figure 7.

The largest sub-group median is 1.77% (additive perturbation method); every other sub-group median sits below 1.5%. The baselines fit just as tightly as the candidate-feature directions: PCA and Random have sub-group median residuals of 1.24% and 0.77%, both below Contrastive's 1.12%.

C. Toy model details

C.1. Architecture

The toy model follows Hänni et al. (2024); Vaintrob (2026). It is a two-layer network with tied weights: an encoder $E \in \mathbb{R}^{H \times d}$ and decoder $D = E^\top$, with a $\tanh^3$ activation function applied element-wise to the hidden layer. The entries of E are drawn i.i.d. from $\{0, +1, -1\}$ with probabilities $\{1 - p, p/2, p/2\}$, where p is a sparsity parameter controlling the fraction of non-zero entries. The network receives a noisy two-hot input $x = x_{\text{clean}} + \epsilon$, where x_{clean} has exactly $s = 2$ active coordinates set to 1 and the remaining set to 0, and $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma^2 = 0.03$. The network operates in superposition, with the number of features d exceeding the number of neurons H.

C.2. Optimization

We do not train the encoder and decoder via gradient descent. Instead, we fix the random encoder structure and optimize two scalar parameters: an input gain c_{in} and an output scale c_{out}, so that the network computes $y = c_{\text{out}} \cdot \tanh^3(c_{\text{in}} \cdot Ex)$.

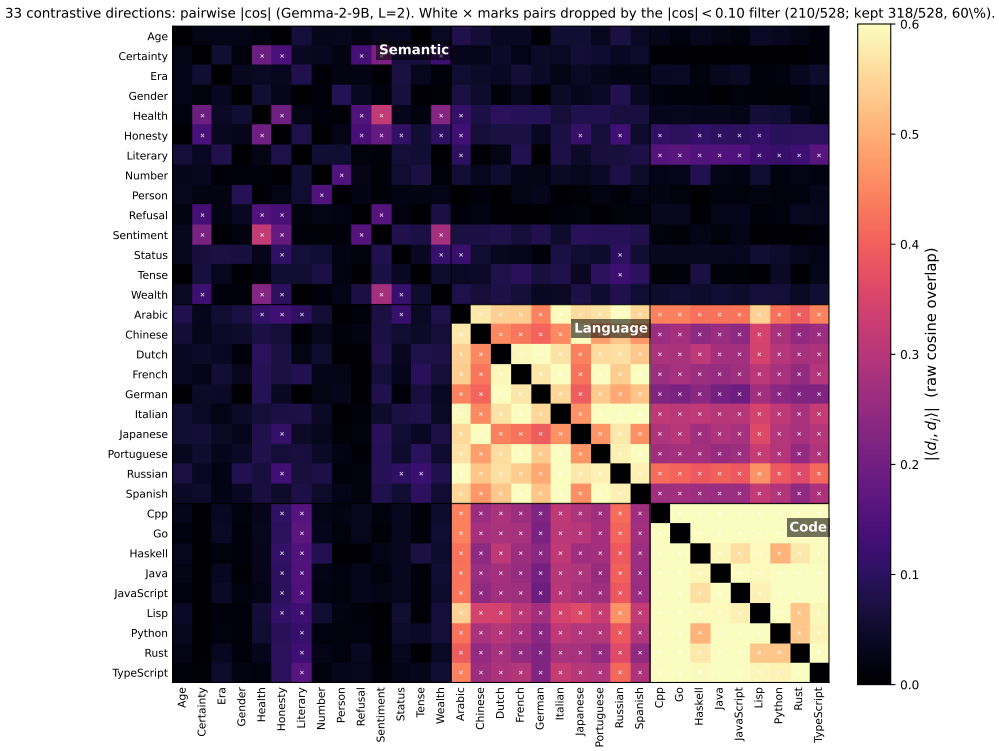


Figure 6. **Pairwise cosine overlap of the 33 contrastive directions.** Cells show $|\langle \mathbf{d}_i, \mathbf{d}_j \rangle|$ on Gemma-2-9B at layer 2. Black gridlines separate the three families (semantic, natural language, programming language). White $\times$ marks the 210 pairs dropped by the $|\cos| < 0.1$ filter applied throughout the paper; the remaining 318 pairs are the ones plotted as Contrastive in the beeswarms. Most filtered pairs sit inside the natural-language block.

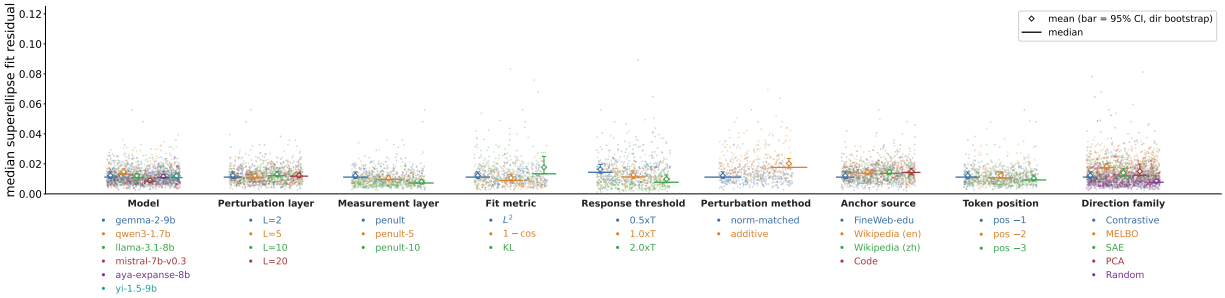


Figure 7. **Distribution of superellipse fit residuals across all conditions.** Each dot is one pair's fit residual ρ. The seven left-most columns mirror the ablation axes of Figure 4 (Model, Perturbation layer, Measurement layer, Fit metric, Response threshold, Perturbation method, Token position, Anchor source). The right-most column gives the residual distribution per direction family (Contrastive, MELBO, SAE, PCA, Random). Per-sub-group mean (white diamond) $\pm$ 95% direction-bootstrap CI and median (horizontal bar) are colour-matched to the sub-group's dots. The y-axis is clipped at 12.5% to keep the bulk of the distribution legible; this clips 2 of the 9,935 plotted dots (<0.05%).

We minimize a per-coordinate weighted MSE loss:

$$\mathcal{L} = \frac{1}{B \cdot d} \left[\lambda \sum_{i \in \text{on}} (c_{\text{out}} f_i - 1)^2 + \sum_{i \in \text{off}} (c_{\text{out}} f_i)^2 \right], \quad (12)$$

where $f = D \tanh^3(c_{\text{in}} \cdot Ex)$, "on" denotes coordinates where $x_{\text{clean}} = 1$, and "off" denotes coordinates where $x_{\text{clean}} = 0$. The weighting λ upweights active-coordinate residuals to discourage the trivial solution of outputting zero, which otherwise dominates for very sparse inputs. Given c_{in}, the optimal c_{out} has a closed-form solution:

$$c^*_{\text{out}} = \frac{\lambda \cdot \langle f, x_{\text{clean}} \rangle_{\text{on}}}{\lambda \cdot \|f\|^2_{\text{on}} + \|f\|^2_{\text{off}}}. \quad (13)$$

We sweep c_{in} over a 121-point grid on $[0.05, 8.0]$ and select the value minimizing $\mathcal{L}$.

The sparsity parameter p was swept over

$\{0.025, 0.05, 0.075, 0.1, 0.125, 0.15\}$ for each (d, H) configuration, selecting p^* by minimum training loss. The selected values were $p^* = 0.15$ for $4\times$, $p^* = 0.10$ for $8\times$, and $p^* = 0.075$ for $16\times$. The weighting parameter was set to $\lambda \in \{500, 1000, 2000\}$ for $d \in \{4096, 8192, 16384\}$ respectively, scaled roughly as $\lambda \approx d/(s \cdot \text{SNR}^2)$ with $s = 2$ and empirical SNR ≈ 2.

C.3. Perturbation procedure

The active set is fixed to $S = \{0, 1\}$ throughout, so all perturbations are performed around the same clean input $x^* = e_0 + e_1$. We sample $N = 80$ pairs (j, k) uniformly without replacement from the inactive coordinates $\{2, 3, \ldots, d-1\}$. For each pair, we perturb along the two ground-truth feature directions e_j and e_k. To measure the effect of misalignment, we rotate e_j and e_k toward random directions by varying amounts. Specifically, for each pair we sample an isotropic random unit pair (w_u, w_v), mutually orthogonal but otherwise unconstrained, and interpolate between the feature direction and the random direction at each misalignment level. The same (w_u, w_v) pair is used across all misalignment levels for a given feature pair.

C.4. Superellipse fitting

For each direction pair and misalignment level, we sweep 60 mixing angles φ uniformly on $[0, \pi/2]$. For each φ, we perform a 2500-point linear scan of the perturbation magnitude α on $[0, 25]$ and identify the smallest α at which $\|f(x^* + \alpha \cdot \hat{d}(\varphi)) - f(x^*)\|_2 > \tau$. The threshold is set to $\tau = 0.5 \cdot \max_{\alpha \le 25} \|f(x^* + \alpha \cdot e_j) - f(x^*)\|_2$ for the first inactive feature $j = 2$, computed per network. The superellipse exponent is estimated by minimizing $\sum_\varphi \left((x_\varphi / r_0)^n + (y_\varphi / r_{\pi/2})^n - 1\right)^2$ over $n \in [0.5, 60]$, where $(x_\varphi, y_\varphi) = \alpha(\varphi) \cdot (\cos\varphi, \sin\varphi)$. The perturbation sweep is deterministic (no noise is added at perturbation time). All random seeds are set to 42. We show in Figure 8 an example of the elliptical boundaries we obtain for the toy model in the style of Figure 1b in the main text.

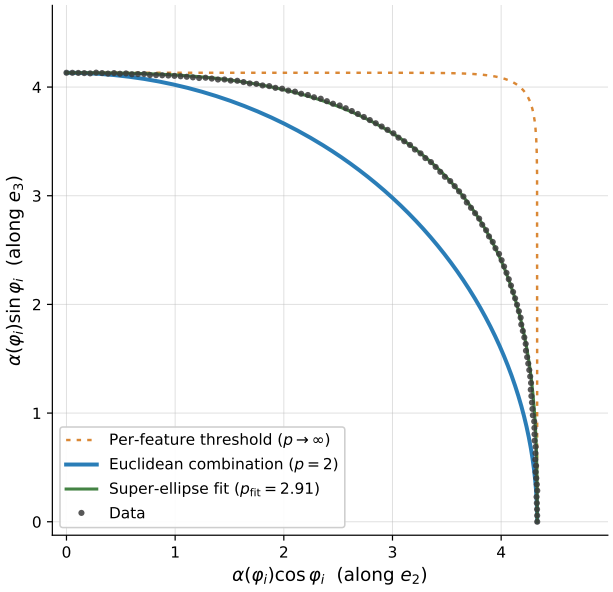


Figure 8. **Iso-plateau boundary.** Plateau-breaking perturbation magnitudes for perturbations aligned with combinations of two feature directions in the toy model with $d = 8192$. The fitted superellipse exponent is 2.91 with a mean residual of 0.13%.